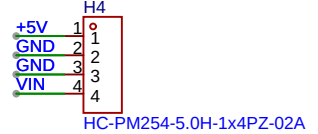
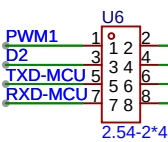
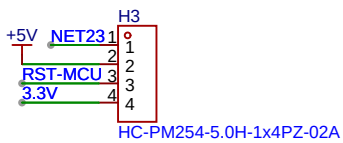
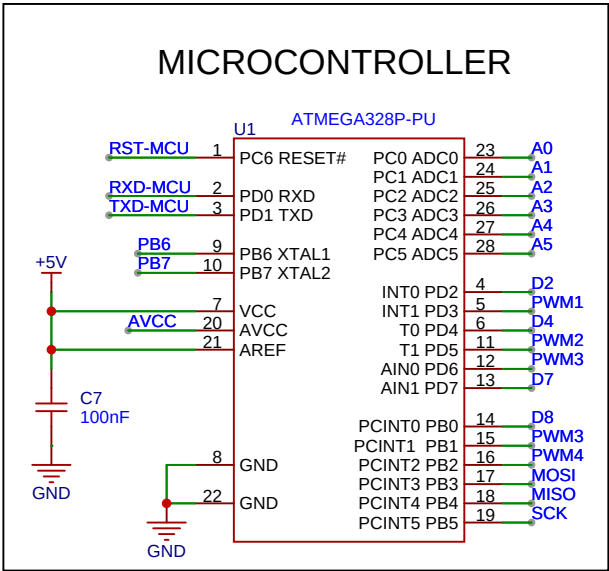
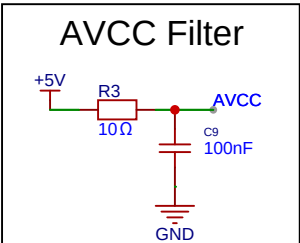
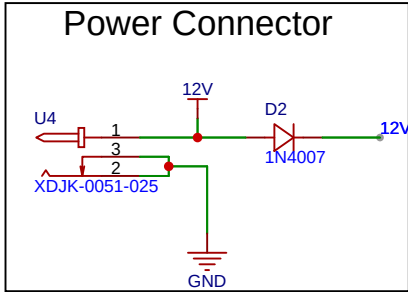
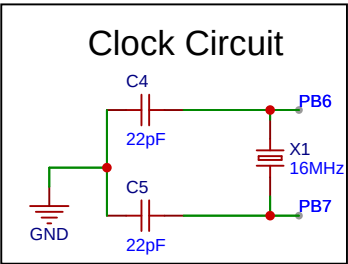
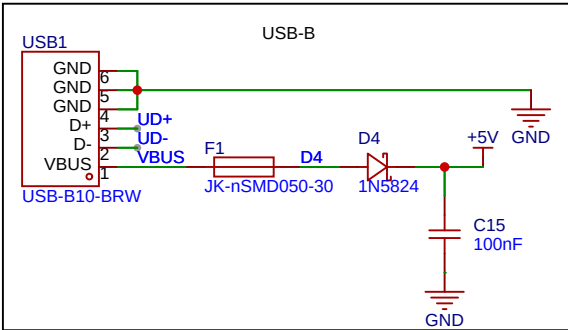
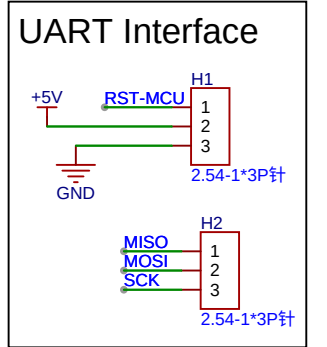
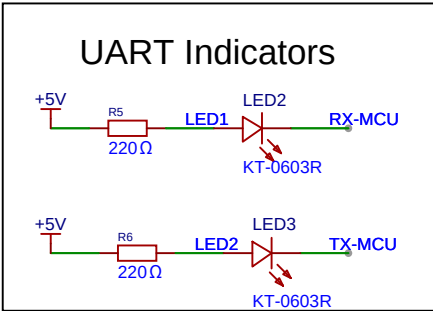
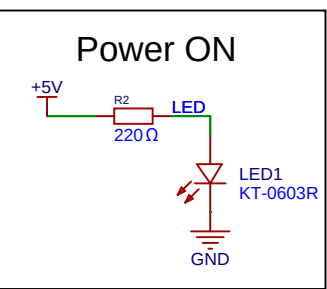
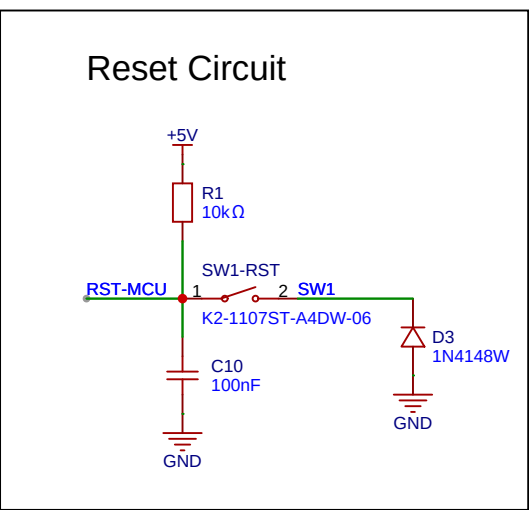
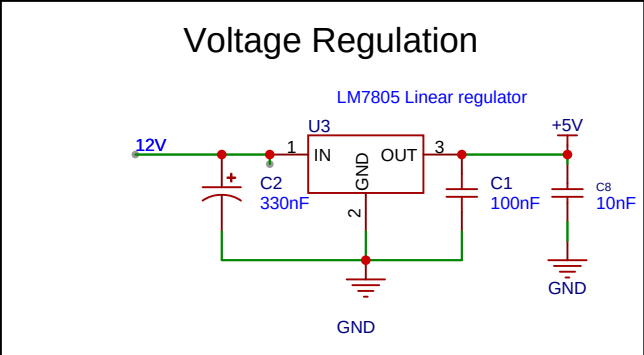
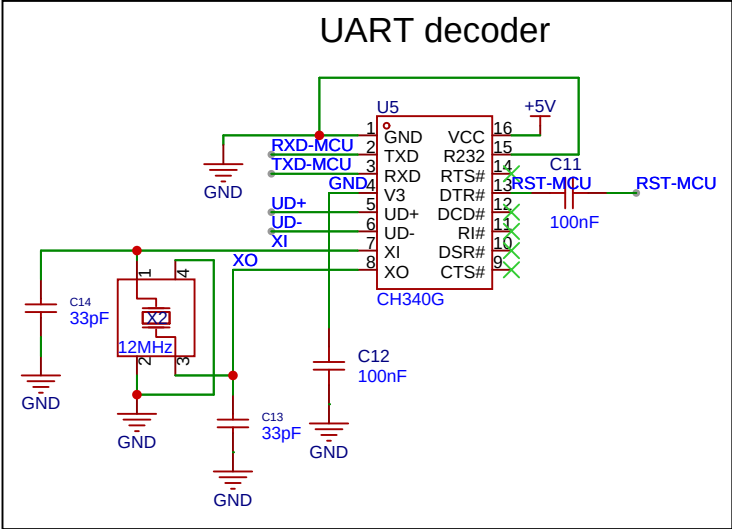


Cover Page

Power Input Stage (12V)	DC Jack or Terminal Block Reverse polarity protection (diode)
Voltage Regulation (12V → 5V)	Choose ONE: Linear Regulator (7805) Buck Converter (bonus)
ATmega328P Core Circuit	VCC, AVCC properly connected All GND pins connected 100nF decoupling capacitors near each VCC AREF capacitor (100nF)
Clock Circuit	16 MHz crystal + 2 capacitors
Reset Circuit	10kΩ pull-up resistor Push button to GND
ICSP Programming Interface	Add 6-pin header: MISO MOSI SCK RESET VCC GND
UART Interface	Add header 4 pins TX (from MCU) RX (to MCU) GND VCC
Status LED	LED + resistor connected to GPIO

Schematic	Schematic1		Update Date	2026-05-02
			Create Date	2026-05-02
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Drawn		Assignment 2-Design single-layer AT mega328p		
Reviewed				
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Schematic	Schematic1			Update Date	2026-05-17
				Create Date	2026-05-02
Page	P2			Part Number	
Drawn		Assignment 2-Design single-layer AT mega328p			
Reviewed					
		VER	SIZE	PAGE	2 OF 2
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